Tooba Imtiaz

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Education

2021 - Apr 2027 (expected) **PhD Candidate,** Electrical Engineering, *Northeastern University*, Boston | **Advisor:** Prof. Jennifer Dy

2018 - 2020 **Masters,** Electrical Engineering, (**GPA**: 3.87/4.3) *Korea Advanced Institute of Science and Technology (KAIST)*, S. Korea | **Advisor:** Prof. In-So Kweon

2014 - 2018 **Bachelors,** Electrical Engineering, (**GPA**: 3.93/4.0, **Rank:** 5th/156) *National University of Sciences and Technology (NUST)*, Pakistan | **Advisor:** Prof. Faisal Shafait

Work Experience

May – Aug 2026 **PhD Intern** | Waymo, Perception Team
Mountain View, CA
- Developed a discrete BEV codebook and Student-Teacher distillation architecture for autonomous perception.
- Diagnosed cross-modal representation bottlenecks and information ceilings to design technical roadmaps for future sensor robustness research.

June – Nov 2025 **Student Researcher** | Google, Pixel Biometrics AI Research (BAIR)
Seattle, WA
- Developed a text-conditioned multi-view latent diffusion model for stylized 3D scene generation.

Sep 2024 – May 2025 **Student Researcher** | Google Beam (Project Starline)
Cambridge, MA
- Continued research as an extension of my internship project for Google Beam.
- Proposed a feed-forward, generalizable 3DGS-based novel view synthesis framework capable of reconstructing wide-coverage, high-resolution scenes, achieving state-of-the-art performance in a single inference pass. ("LVT", SIGGRAPH Asia 2025)

May – Aug 2024 **Research Intern** | Google Beam (Project Starline)
Playa Vista, CA
- Worked on efficient 3D reconstruction and novel-view synthesis for Google Beam.
- Proposed feed-forward architecture achieved plausible quantitative and qualitative results, despite its simplicity.

Sep 2021 – Present **Research Assistant** | Machine Learning Lab @ SPIRAL,
Northeastern University, Boston
- Developed a novel view synthesis framework to visualize previously unseen depths in 3D reflectance confocal microscopy (RCM) images of human skin, to facilitate early detection of skin diseases and cancers. ("CD-RCM", NeurIPS 2026)
- Member of the Northeastern PAIR (Physical AI Research) initiative for developing a physics-aware world model
- Contributed to a regularization-based approach for improving continual learning. ("STAR", ICLR 2025)
- Developed an optimization-based sparse adversarial attack on images and evaluated its interpretability. ("SAIF", TMLR 2025)
- Implemented NeRF-based 3D scene reconstruction from phone camera videos to facilitate at-home patient health monitoring.
- Formulated a clustering-based loss to improve the performance of 3D-object detection from multiview 2D inputs.

Spring 2023 **Teaching Assistant** | EECE7397 Advanced Machine Learning, Northeastern University

Sep 2020 - Aug 2021 **Consultant - ML and AI** | Endress+Hauser,
Maulburg, Germany
Proposed ML and CV-based solutions for process automation and optimization. Led two projects, both deployed to production:
- **Deep learning for unsupervised 3D classification:** used Autoencoders, Capsule architectures, and Implicit Neural Networks.
- **Forecasting on time series:** utilized DNNs and Temporal Transformers to predict compound concentrations in liquids using sensors measuring base physical quantities. Achieved $\sim 96\%$ accuracy w.r.t. specialized physical sensors.

Sep 2018 - Aug 2020 **Research Assistant** | Robotics and Computer Vision Lab,
KAIST, South Korea
- **Bosch-RCV Project:** Performed camera calibration, data collection, and vehicle trajectory estimation. Designed an occlusion-robust vehicle re-identification method using GANs for seamless tracking across a multi-camera surrounding awareness system.
- **Universal Adversarial Perturbations:** Developed novel adversarial attack algorithms. Published at CVPR, AAAI, and ACCV '20.

Sep 2015 - May 2018 **Research Intern** | TUKL-NUST R&D Centre,
NUST, Pakistan
- Proposed table detection and parsing in document images using ML and CV (LSTMs, text classification, clustering algorithms).
- Implemented LSTMs for handwritten address recognition to sort postal mail.

Publications

CD-RCM: Generalizable Continuous-Depth Novel View Synthesis for Reflectance Confocal Microscopy | **NeurIPS 2026**
T. Imtiaz, M. Rajadhyaksha, K. Kose, J. Dy

PanoWorld: Geometry-Consistent Panoramic Video World Modeling | **NeurIPS 2026**
L. Jiang, X. Bai, B. Galoaa, S. Moezzi, C. J. Lee, **T. Imtiaz**, E. Yeh, J. Dy, Y. Wang, S. Ostadabbas

Human Cognition in Machines: A Unified Perspective of World Models | **Preprint 2026**
T. Rupprecht*, P. Zhao*, A. Taherin*, A. Akbari*, A. Akbari*, Y. He*, **T. Imtiaz***, et al.

LVT: Large-Scale Scene Reconstruction via Local View Transformers | **SIGGRAPH Asia 2025**
T. Imtiaz*, L. Chai*, K. Heal, X. Luo, J. Park, J. Dy, J. Flynn

STAR: Stability-Inducing Weight Perturbation for Continual Learning | **ICLR 2025**
M. Eskander, **T. Imtiaz**, D. Hill, Z. Wang, J. Dy

ADAPT to Robustify Prompt Tuning Vision Transformers | **TMLR 2025**
M. Eskander, **T. Imtiaz**, Z. Wang, J. Dy

SAIF: Sparse Adversarial and Imperceptible Attack Framework | **TMLR 2025**
T. Imtiaz, M. Kohler, J. Miller, Z. Wang, M. Eskandar, M. Sznaier, O. Camps, J. Dy

Volumetric Propagation Network: Stereo-LiDAR Fusion for Long-Range Depth Estimation | **IEEE RA-L 2021**
J. Choe, K. Joo, **T. Imtiaz**, I.S. Kweon

Understanding Adversarial Examples from the Mutual Influence of Images and Perturbations | **CVPR 2020**
C. Zhang*, P. Benz*, **T. Imtiaz**, I.S. Kweon

CD-UAP: Class Discriminative Universal Adversarial Perturbation | **AAAI 2020**
C. Zhang*, P. Benz*, **T. Imtiaz**, I.S. Kweon

Double Targeted Universal Adversarial Perturbations | **ACCV 2020**
P. Benz*, C. Zhang*, **T. Imtiaz**, I.S. Kweon

Data from Model: Extracting Data from Non-robust and Robust Models | **CVPRW 2020**
P. Benz*, C. Zhang*, **T. Imtiaz**, I.S. Kweon

Universal Adversarial Perturbations are Not Bugs, They are Features | **CVPRW 2020**
P. Benz*, C. Zhang*, **T. Imtiaz**, I.S. Kweon

Patents

3D Novel View Synthesis from Sparse Optical Sectioning Images | **Pending**
T. Imtiaz, K. Kose, J. Dy, M. Rajadhyaksha
A feedforward 3D synthesis method for generating continuous tissue views from sparse optical sections.

LVT: Large-Scale Scene Reconstruction via Local View Transformers | **Pending**
J. Flynn, L. Chai, **T. Imtiaz**
A generalizable novel view synthesis framework for reconstructing wide-coverage, high-resolution scenes, achieving state-of-the-art performance.

Invited Talks

Jun 2026	Voxel51: AI, ML & CV Meetup: Presented my research work (LVT) on generalizable, feed-forward novel view synthesis for large-scale scenes.
Jul 2019	The 2nd KEEP-I (KAIST EE Partners - International): Presented my research work at KAIST to ∼60 invited professors from existing and prospective KAIST partner universities.
Sep 2019	KAIST EE Visit Camp: Delivered an invited talk reflecting on my academic and overall experience at KAIST.

Scholarships and Awards

2020	Qualcomm Innovation Fellowship Award, South Korea (among the 20 awardees)
2018-20	KAIST Scholarship: Awarded on merit
2018	KAIST-EE Visit Camp: Among the 50 students selected and sponsored from 10 countries.
2014-2018	NUST Merit Scholarship (Awarded to top-3 GPA holders of cohort)
2017	Global UGRAD Exchange Program, US Dept of State (∼ 7.6% selection rate)

Skills

Python	JAX, PyTorch, Tensorflow, Keras, Numpy, scikit-learn, cuda, Matplotlib
C / C++ / Java	Object-oriented programming, Data structures, frontend and backend dev
Misc.	MATLAB, Unix, gcc, Git, SQL, LaTeX, ROS, AutoCAD

Academic Service

Workflow Chair	AAAI 2024 - Managed the AAAI 2024 paper review process for 12,100 submissions, working with 7k reviewers, 765 senior program committee (SPC), and 320 area chairs (AC). - Used topic modeling and text similarity to determine reviewer, SPC, and AC assignments.
Conference Reviewer	NeurIPS 2025-26, ECCV 2024 & 26, ICLR 2025-26, AAAI 2026, CVPR 2024, ACCV 2024, ICCV 2023.
Volunteer	NeurIPS 2025, ICML 2022.